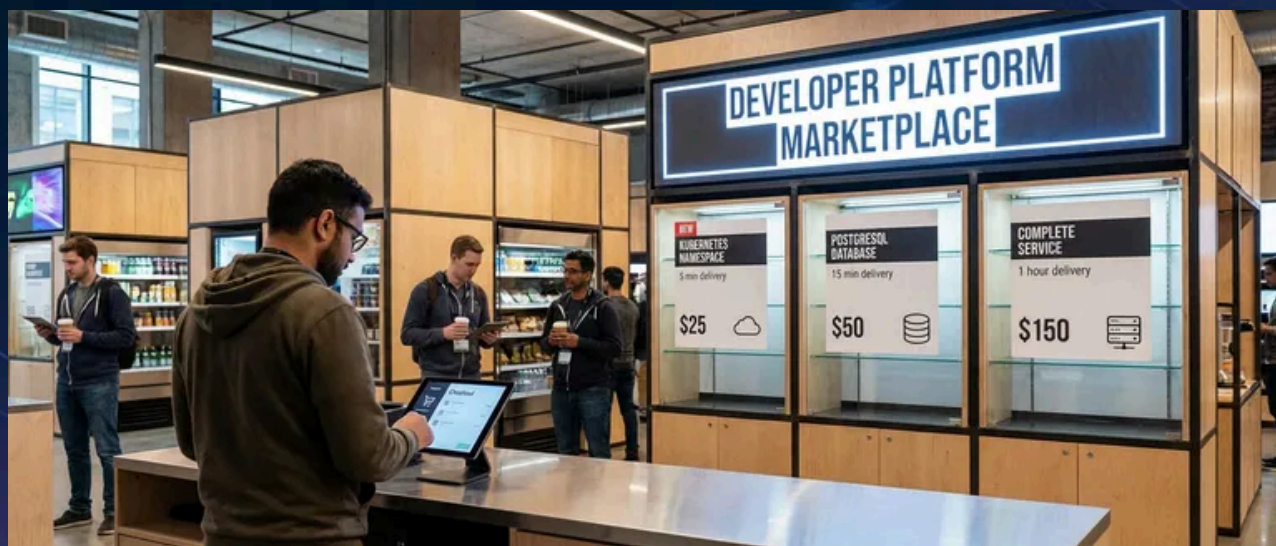


Portal vs Platform: Why Your IDP Is Just Links



Published on February 6, 2022



Webstack
Builders

Table of Contents

The Portal Maturity Model 3

Self-Service Action Design 5

 Anatomy of a Good Action 5

 Action Design Guidelines 7

Workflow Automation 8

 The Anatomy of a Multi-Step Workflow 8

 Designing for Failure 9

Approval Workflows 10

 Approval Policy Dimensions 10

 Making Approvals Fast 11

Integration Patterns 12

 Integration Categories 12

 The Adapter Pattern 13

 Metric Categories 14

 Interpreting Metric Combinations 15

 The Metrics That Matter Most 16

Conclusion 16



Last month I watched a developer spend three days getting a PostgreSQL database. The company had a beautiful internal developer portal - Backstage, fully themed, comprehensive service catalog, scorecards showing tech debt metrics. The developer found the “Database Provisioning” page in under 30 seconds. Then he read the instructions: file a Jira ticket with the DBA team, wait for triage, attend a 15-minute review meeting to justify the request, and receive credentials via a separate ticket once provisioning completed. Three to five business days, assuming no holidays or sick leave on the DBA team.

The portal worked perfectly. The developer knew exactly where to go and what to do. And yet the outcome was identical to not having a portal at all - the same tickets, the same meetings, the same waiting.

That’s the portal-versus-platform distinction that trips up most organizations. A portal *indexes* things: it aggregates links, surfaces documentation, shows who owns what. A platform *does* things: it provisions infrastructure, creates repositories, grants access. The portal tells you how to request a database. The platform gives you a database.

WARNING

If your “platform” requires developers to read documentation and file tickets to get things done, you’ve built a directory, not a platform. Platforms execute workflows. Directories point to workflows that humans execute.

The difference matters because it determines whether your internal developer platform actually reduces toil or just makes toil easier to find. Self-service actions are what transform a directory into a platform. Without them, you’ve built a prettier wiki.

The Portal Maturity Model

Most internal developer portals evolve through predictable stages. Understanding where you are helps clarify where to invest next - and where most organizations get stuck.

➤ Level 1: Directory.

This is the starting point - a central place that aggregates links to documentation, team contacts, and runbooks. Think Confluence home pages, GitHub wikis with service lists, or spreadsheets tracking ownership. The value is real: developers know where to look. But they still have to do everything manually. Finding the database provisioning doc doesn't provision the database.



➤ **Level 2: Search.**

The directory becomes searchable and indexed. Backstage with a service catalog lives here, along with API registries and dependency graph visualizations. Discovery is faster. Context switching drops. But finding something quickly doesn't mean you can act on it quickly. The developer who locates the database docs in 30 seconds still files the same ticket and waits the same 3-5 days.

➤ **Level 3: Insights.**

Now the portal shows health metrics, ownership data, and quality scorecards. You can see which services have test coverage gaps, which teams own what, and how much each service costs. Visibility improves dramatically. But seeing problems doesn't fix them. The dashboard showing your DBA team has a 47-ticket backlog doesn't get your database provisioned faster.

➤ **Level 4: Platform.**

Self-service actions with automation. Developers can provision databases, create services, and request access - not by finding documentation, but by clicking buttons and filling forms that trigger real workflows. The difference isn't incremental. It's the difference between "3-5 business days" and "15 minutes."

Level	What Developers Can Do	What They Still Can't Do
Directory	Find documentation	Execute workflows
Search	Quickly find services and APIs	Provision or modify resources
Insights	See health and ownership	Fix issues without tickets
Platform	Self-service common tasks	Only edge cases need tickets

Portal maturity levels.

INFO

Most organizations plateau at Level 2 or 3. They build beautiful service catalogs with scorecards and graphs, but developers still file tickets for everything. The jump to Level 4 requires different thinking: instead of "what should developers know?" ask "what should developers be able to do?"



Self-Service Action Design

The difference between a good self-service action and a frustrating one often comes down to design decisions that seem minor but compound. A database provisioning action that takes 30 seconds to fill out and 15 minutes to complete feels magical. The same action with a confusing form, unclear outcomes, and no progress visibility feels like a different kind of ticket system.

Anatomy of a Good Action

A well-designed action has clear structure: identity (what it is), inputs (what the user provides), workflow (what happens), and outputs (what the user gets back). Here's what that looks like for database provisioning:

backstage-action-definition.yaml

```

1  # Backstage scaffolder action for PostgreSQL provisioning
2  apiVersion: scaffolder.backstage.io/v1beta3
3  kind: Template
4  metadata:
5    name: provision-postgresql-database
6    title: Create PostgreSQL Database
7    description: Provision a new PostgreSQL database with configured access
8    tags: ["postgresql", "database", "self-service"]
9  spec:
10   owner: platform-data-team
11   type: database
12
13   parameters:
14     - title: Database Configuration
15       required: [name, environment, size]
16       properties:
17         name:
18           title: Database Name
19           type: string
20           pattern: "^[a-z][a-z0-9-]{2,30}$"
21           description: Lowercase letters, numbers, hyphens. 3-31 chars.
22         environment:
23           title: Environment
24           type: string
25           enum: [dev, staging, production]
26           default: dev

```



```

27     size:
28       title: Instance Size
29       type: string
30       enum: [small, medium, large]
31       enumNames:
32         - Small (2 vCPU, 8GB) - $50/mo
33         - Medium (4 vCPU, 16GB) - $150/mo
34         - Large (8 vCPU, 32GB) - $400/mo
35       default: small
36
37   steps:
38     - id: create-terraform-pr
39       name: Create Infrastructure PR
40       action: publish:github:pull-request
41       input:
42         repoUrl: github.com?repo=infrastructure&owner=myorg
43         branchName: db-{{ parameters.name }}
44         title: "Provision database: {{ parameters.name }}"
45
46     - id: wait-for-apply
47       name: Apply Infrastructure
48       action: terraform:wait-for-apply
49       input:
50         prUrl: {{ steps.create-terraform-pr.output.remoteUrl }}
51
52     - id: store-credentials
53       name: Store Credentials
54       action: vault:create-secret
55       input:
56         path: databases/{{ parameters.name }}

```

Backstage scaffolder action for database provisioning.

The YAML is straightforward, but the design decisions matter. Notice the sensible defaults (dev environment, small size), the clear cost indication in the enum labels, and the validation pattern that prevents naming mistakes before submission. For more on Backstage scaffolder actions, see the Backstage documentation on software templates <<https://backstage.io/docs/features/software-templates/>> .



Action Design Guidelines

1

Obvious outcomes.

Users should know exactly what will happen before clicking submit. Show a preview of resources to be created, display cost estimates, list permissions that will be granted. The anti-pattern is "submit and pray" - the user clicks a button and has no idea what happens next.

2

Sensible defaults.

Most fields should have good defaults for the common case. Pre-fill the team from user context. Default to the smallest viable size. Default to dev environment. A blank form requiring research to fill out is a friction multiplier.

3

Guardrails, not gates.

Prevent mistakes without blocking legitimate work. Validate inputs in real-time. Warn about expensive choices. But auto-approve within policy bounds - if every request requires manual approval, you haven't built self-service, you've built a ticket system with a nicer UI.

4

Transparent execution.

Users can see what's happening and why. Show workflow steps and progress. Link to the underlying PR or pipeline. Explain approval requirements when they apply. Surface errors clearly with remediation steps.

5

Recoverable failures.

Failures should be fixable without starting over. Allow retrying failed steps. Let users edit and resubmit. Provide clear error messages with remediation guidance. If a failure requires filing a support ticket, the action has failed its purpose.

✓ SUCCESS

The best self-service actions feel like filling out a form to order something online: clear options, sensible defaults, transparent pricing, and a tracking number when you submit. If your action feels like filling out a government form, redesign it.



Workflow Automation

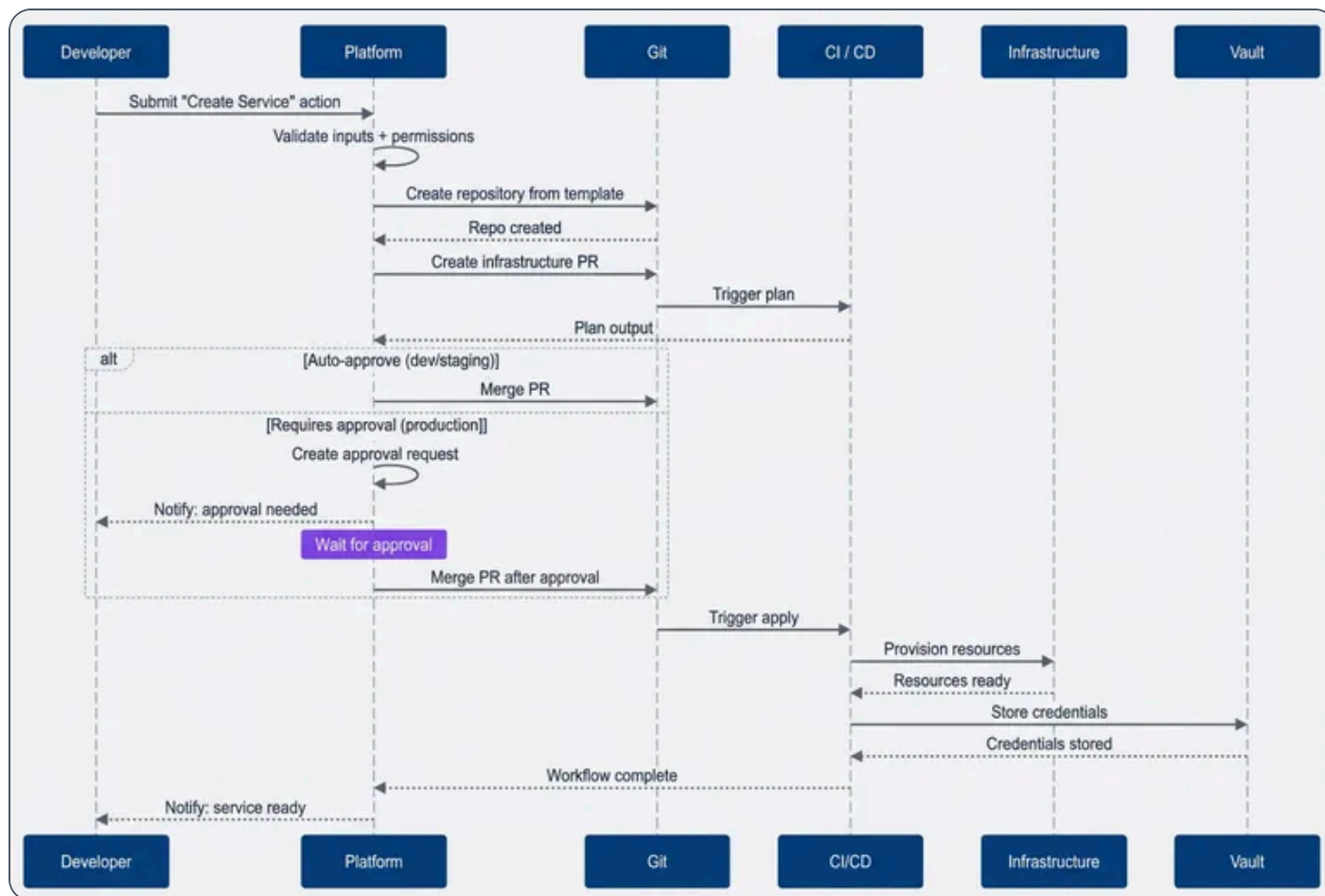
Self-service actions rarely involve a single step. Provisioning a new service might mean creating a repository, generating infrastructure-as-code, waiting for Terraform to apply, storing credentials in Vault, and registering the service in the catalog. Each step can fail. Each step depends on outputs from previous steps. And if something fails halfway through, you need a plan for the orphaned resources.

This is workflow orchestration - coordinating multi-step processes with proper error handling, progress visibility, and rollback support.

The Anatomy of a Multi-Step Workflow

Consider what happens when a developer requests a new service. The portal validates their inputs and permissions, creates a repository from a template, generates a Terraform PR (Pull Request) for the service's infrastructure, waits for approval (or auto-approves for non-production), applies the Terraform changes, stores the resulting credentials, and notifies the developer. That's seven steps with dependencies between them.





Multi-step service creation workflow.

The sequence diagram shows the happy path, but real workflows need to handle failures at every step. What if the repository creation succeeds but the Terraform PR (Pull Request) fails? What if the apply times out? What if credential storage fails after infrastructure is already provisioned?

Designing for Failure

Each workflow step needs three things: a clear action, defined failure behavior, and (where appropriate) a rollback action.

Failure behaviors fall into a few categories. Some failures should abort the entire workflow - if validation fails, there's no point continuing. Some should retry - transient API errors often resolve on the second attempt. Some should skip - failing to send a notification shouldn't block an otherwise successful provisioning. And some should trigger manual intervention - edge cases that automation can't handle.



Rollback steps clean up resources created by earlier steps when a later step fails. If you create a repository in step 2 but the infrastructure PR (Pull Request) fails in step 3, the rollback should delete that repository. Without rollback, failed workflows leave orphaned resources that accumulate over time - repositories nobody uses, partial infrastructure deployments, dangling secrets.

The key insight is that **non-critical steps should fail gracefully**. Registering a service in the catalog is nice to have, but if it fails, the service still exists and works. Mark these steps as “skip on failure” so they don’t block the overall workflow.

⚠ WARNING

Every workflow needs a rollback plan. If step 4 fails, what happens to resources created in steps 1-3? Orphaned repositories, half-provisioned infrastructure, and dangling credentials erode trust in self-service. Design for failure from the start.

Approval Workflows

Approvals are where self-service platforms often regress into ticket systems. The instinct is understandable - production changes are risky, expensive resources need oversight, compliance requires audit trails. But if every request routes through a manual approval queue, you’ve just built Jira with a different skin.

The solution is **policy-based approval routing**: define rules that auto-approve low-risk requests while routing high-risk ones to the right approvers. Most requests should flow through automatically. Approvals become the exception, not the rule.

Approval Policy Dimensions

Different situations warrant different approval requirements. Here are the dimensions that typically matter:

Environment-based:

Dev and staging requests auto-approve instantly. Production requests require team lead approval. This is the most common pattern and handles the majority of cases.



Cost-based:

Resources under \$100/month auto-approve. Resources between \$100-500 need team lead sign-off. Resources over \$500 require team lead plus finance approval. Thresholds vary by organization, but the principle is consistent: more expensive means more oversight.

Scope-based:

Changes affecting only your own team auto-approve. Changes affecting multiple teams require approval from each affected team's lead. This prevents one team from unilaterally impacting another's infrastructure.

Risk-based:

Standard operations auto-approve. High-risk operations - deleting databases, modifying IAM permissions, changing network configs, rotating encryption keys - require security review regardless of environment.

Time-based:

Normal business hours follow standard policies. Change freeze windows require explicit exception approval. This prevents well-meaning developers from deploying during critical periods.

These dimensions combine. A production database deletion during change freeze might require team lead approval **and** security review **and** change freeze exception - three separate approvers. A dev environment S3 bucket during business hours? Instant auto-approval.

In Backstage, you'd implement these rules through a custom approval plugin or integrate with Open Policy Agent (OPA) for policy-as-code. The scaffolder evaluates the policy before executing steps, routing to the appropriate approval workflow or proceeding automatically based on the policy decision.

Approval Type	Use Case	SLA	Auto-Escalation
Auto-approve	Dev/staging, low cost	Instant	N/A
Multiple approvers	High risk, multi-team	8 hours	After 4 hours
Sequential approval	Compliance-critical	24 hours	After 8 hours

Approval types and SLAs.



Making Approvals Fast

Even when manual approval is required, it shouldn't be slow. A few practices help:

➤ **Give approvers context.**

The approval notification should include everything needed to make a decision: what's being requested, who's requesting it, what environment, estimated cost, risk level, and links to the Terraform plan or other details. Approvers shouldn't need to dig through systems to understand the request.

➤ **Set SLAs with auto-escalation.**

A production approval that sits in queue for 6 hours is blocking a developer. Set expectations (e.g., 4-hour SLA for standard production requests) and auto-escalate to backup approvers when SLAs are at risk.

➤ **Enable approval from notifications.**

If your platform sends Slack notifications, include approve/reject buttons in the message. If it sends emails, include one-click approval links. Every click required to approve is friction that slows the process.

➤ **Track approval metrics.**

Measure time-to-approval by policy type. If production approvals consistently take 8 hours when the SLA is 4, you either need more approvers, faster approvers, or a policy change.

INFO

Approvals should be the exception, not the rule. If most requests require manual approval, you haven't built self-service - you've built a ticket system with a nicer UI. Use policy-based auto-approval aggressively for low-risk operations.

Integration Patterns

A self-service platform doesn't exist in isolation. It needs to talk to your Git provider to create repositories, your infrastructure tooling to provision resources, your secrets manager to store credentials, your observability stack to set up monitoring, and your communication tools to send notifications. The integration layer is where the platform connects to reality.

Integration Categories

Most platforms need integrations across five categories:



1

Git providers

Handles repository creation, pull request management, and team permissions. GitHub and GitLab are the common choices. The platform creates repos from templates, opens PRs for infrastructure changes, and manages who can access what. Authentication typically uses GitHub Apps or access tokens with appropriate scopes.

2

Infrastructure provisioning

Where resources actually get created. Terraform Cloud (or Atlantis for self-hosted) handles the IaC workflow - creating workspaces, queuing plans and applies, reading state outputs. For direct resource creation, you need AWS / GCP / Azure SDK access via IAM roles or service accounts. Kubernetes clusters need ServiceAccount tokens for applying manifests and managing RBAC.

3

Secrets management

Stores and distributes credentials securely. HashiCorp Vault is the most common choice, using AppRole authentication for the platform to create secrets and generate dynamic credentials. AWS Secrets Manager is simpler if you're already AWS-native.

4

Observability

Integration means the platform can provision monitoring alongside infrastructure. Datadog or Prometheus for metrics, PagerDuty for alerting. When a new service is created, its dashboards and alerts should be created automatically - not added manually weeks later when someone remembers.

5

Communication

Enables notifications and interactive approvals. Slack integration with a bot token lets the platform send messages, create channels, and handle approval buttons directly in chat. Email provides fallback notifications.

The Adapter Pattern

The key architectural insight is that actions shouldn't know which specific provider they're using. A "create repository" step should work whether you use GitHub, GitLab, or Bitbucket. This means designing integrations as adapters with standard interfaces.



Each adapter declares its capabilities (what actions it can perform), implements a standard execution interface, and handles provider-specific authentication. The platform's action definitions reference abstract capabilities like `createRepository(name, template, visibility)` and `createPullRequest(repo, branch, title, body)` rather than provider-specific APIs. The GitHub adapter implements these by calling the GitHub API; the GitLab adapter calls the GitLab API. The action definition doesn't know or care which one runs.

This abstraction pays off when you need to swap providers. Moving from GitHub to GitLab, or from Terraform Cloud to Atlantis, becomes a configuration change rather than a rewrite of every action that touches those systems.

Integration Type	Common Providers	Authentication	Key Capabilities
Git provider	GitHub, GitLab	GitHub App, Access Token	Repos, PRs, permissions
Infrastructure	Terraform Cloud, AWS	API Token, IAM Role	Provision, state, outputs
Secrets	Vault, AWS Secrets Manager	AppRole, IAM Role	Create, rotate, policies
Observability	Datadog, Prometheus	API Key	Dashboards, monitors, SLOs

Common integration patterns.

✓ SUCCESS

Design integrations as adapters with standard interfaces. This lets you swap providers (GitHub to GitLab, Terraform Cloud to Atlantis) without rewriting actions. The action definition shouldn't care which Git provider you use.

Building a platform is an investment - in tooling, integrations, and ongoing maintenance. Justifying that investment requires metrics that show the platform is actually delivering value. The trap is measuring what's easy (action count) instead of what matters (developer productivity).



Metric Categories

Adoption metrics Tell you whether developers are actually using the platform. Track the number of self-service actions executed, the percentage of the engineering org using actions, and the coverage of common tasks available as actions. Growing adoption means the platform is solving real problems. Flat or declining adoption means something's wrong - either discoverability, trust, or the actions themselves.	
Efficiency metrics Measure time savings. Compare time-to-provision before and after the platform: how long did it take to get a database via tickets versus via self-service? Track approval wait times. Measure what percentage of requests auto-approve versus requiring manual intervention. The goal is dramatic improvement - not 20% faster, but 95% faster.	
Quality metrics Track whether the platform is reliable. Action success rate should be above 95% - if actions fail frequently, developers stop trusting them. Rollback rate indicates how often workflows fail partway through and need cleanup. Support ticket reduction shows whether the platform is actually offloading work from the platform team or just creating different work.	
Satisfaction metrics Capture developer perception. Net Promoter Score for the platform overall, ratings for individual actions, and satisfaction when escalation is needed. High efficiency with low satisfaction means the platform works but feels bad to use - maybe the UX is confusing or errors are frustrating.	

Interpreting Metric Combinations

Metrics are most useful in combination. Here's what different patterns indicate:

High efficiency, low adoption

The platform works but developers don't trust it or can't find it. Improve discoverability, add more actions for common tasks, or investigate why developers prefer the old way.

High adoption, low efficiency

Developers use the platform but it's slow. Actions need optimization, approval policies are too strict, or workflows have bottlenecks.



**High adoption, high efficiency,
low satisfaction**

The platform works and is fast, but developers don't like using it. UX problems, confusing errors, or missing features they expected.

**Declining adoption with stable
efficiency**

Something changed. Maybe a competing tool emerged, a key action broke, or the platform team shipped something unpopular. Investigate quickly.

The Metrics That Matter Most

If you can only track a few things, focus on these:

Metric	Baseline (Portal)	Target (Platform)	Value
Time to new database	3-5 days	15 minutes	99% reduction
Time to new service	1-2 weeks	1 hour	95% reduction
Platform tickets/week	50	15	70% reduction
Manual approvals needed	90%	25%	72% reduction

Portal vs platform value comparison.

The first two metrics (time to provision) are the most compelling for stakeholders. “We reduced database provisioning from 3-5 days to 15 minutes” is a story everyone understands. The last two metrics (ticket reduction, auto-approval rate) show operational efficiency - the platform team can do more with less manual intervention.

INFO

Track both efficiency metrics (time saved) and adoption metrics (who's using it). High efficiency with low adoption means developers don't trust the platform. High adoption with low efficiency means actions need optimization. You need both.



Conclusion

The distinction between portal and platform comes down to one question: what happens after a developer finds what they need? In a portal, they read documentation and file tickets. In a platform, they click a button and get resources.

That developer waiting three days for a database? In a real platform, they'd have filled out a form, seen a preview of what would be created, clicked submit, and had credentials in their hands within 15 minutes. No tickets. No meetings. No waiting for another team's availability.

Getting there requires intentional design. Actions need obvious outcomes and sensible defaults. Approval policies should auto-approve aggressively and only route exceptions to humans. Workflow orchestration must handle failures gracefully and clean up after itself. Integrations should abstract away provider specifics so you can evolve the underlying systems without rewriting everything.

✓ SUCCESS

The test of a real platform: can a new developer join, need a database, and have it provisioned without asking anyone for help? If yes, you've built a platform. If no - if they need to read Confluence, file a Jira ticket, or ping someone on Slack - you've built a prettier ticket system.

Level 4 platform maturity isn't cheap. It requires investment in tooling, integration work, and ongoing maintenance. But the returns are proportional: developer productivity multiplied across the organization, platform team leverage instead of ticket-by-ticket toil, and the compounding benefits of standardization and automation.



Copyright © 2022 Webstack Builders, Inc.

The text, diagrams, and images in this work are licensed under CC BY-NC 4.0

All code samples in this article are licensed under the MIT License. Feel free to use, modify, and distribute them in any project.

<https://www.webstackbuilders.com/articles/internal-developer-portal-platform-self-service-actions>

